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MTH634 Topology

Question No : 3 of 33

Which of the following statement is false?

Solved by Moin AKhtar

BCJ Moin Akhtar

Answer (Please select your correct option)

- The set $\mathbb{R}$ with usual topology is metrizable space.
- The set $\mathbb{R}^2$ with usual topology is metrizable space.
- Indiscrete topology on a set X containing more than one point is metrizable

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Question No : 8 of 33

Which of the following statement is true?

Answer (Please select your correct option)

- Every separable space is second countable.
- Every second countable space is separable.
- $\mathbb{R}$ with lower limit topology is second countable.

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MTH634 Topology

Question No : 10 of 33

A topological space is a T_1 – space if and only if its each finite subset is

Answer (Please select your correct option)

- a perfect set
- a closed set
- an open set
- none

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Done

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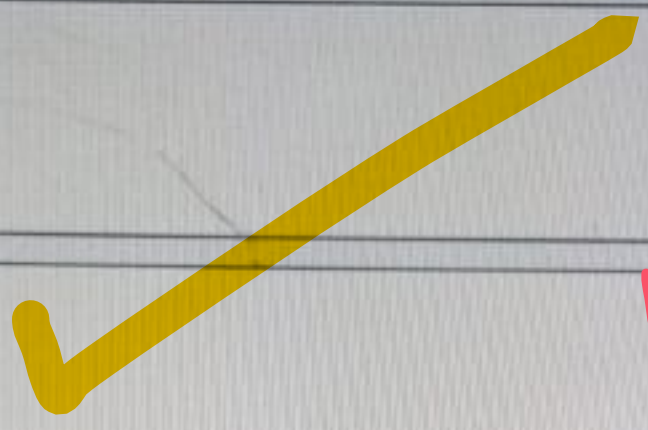
MTH634 Topology

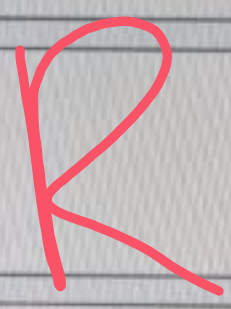
Question No : 11 of 33

A finite set has _____ in a T_1 space.

Answer (Please select your correct option)

- no isolated points
- no limit points
- infinte limit points
- none





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Done

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MTH634 Topology

Question No : 2 of 33

Which of the following statement is true?

Answer (Please select your correct option)

- If a map $f : X \to Y$ is bicontinuos then it is also bijective.
- If a map $f : X \to Y$ is bicontinuos then it is not bijective.
- If a map $f : X \to Y$ is bicontinuos then it may or may not be bijective.

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Done

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MTH634 Topology

Question No : 9 of 33

Which of the following statement is true?

Answer (Please select your correct option)

- A separable metric space is second countable.
- A separable metric space is first countable.
- Every metric space is second countable.
- none

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Done

Start | VU Examination Syste... | Document1 - Microsoft ...

Question No : 6 of 33 Marks: 1 (Bu

Let $B_1(x,r_1)$ and $B_2(x,r_2)$ be two open balls centered at x with radii r_1 and r_2 respectively. Then the following statement is true.

Answer (Please select your correct option)

- $B_1(x,r_1) = B_2(x,r_2)$
- either $B_1(x,r_1) \subset B_2(x,r_2)$ or $B_1(x,r_1) \supset B_2(x,r_2)$
- none

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Question No : 5 of 33

For metric d on X, an open ball of radius r with center $x \in X$ is defined as:

Answer (Please select your correct option)

- $B(x,r) = \{y \in X \mid d(x,y) \leq r\}$
- $B(x,r) = \{y \in X \mid d(x,y) > r\}$
- $B(x,r) = \{y \in X \mid d(x,y) < r\}$
- none

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MTH634 Topology

Question No : 4 of 33

Which of the following statement is false?

Answer (Please select your correct option)

- Discrete topology on a countable set X is second countable
- Any finite set with any topology is second countable
- Discrete topology on a real line R is second countable ✓ R
- The set R with usual topology is second countable

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MTH634 Topology

Question No : 1 of 33

Which of the following statement is true?

Answer (Please select your correct option)

- Projection map may or may not be closed.
- Projection map is open.
- Projection map is closed.

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MTH634 Topology

Question No : 24 of 33 M

If (X,τ) be a path connected topological space, then (X,τ) is connected

Answer (Please select your correct option)

- True
- False

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MTH634 Topology

Question No : 25 of 33

The image of connected topological space is path connected.

Answer (Please select your correct option)

- True
- False

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MTH634 Topology

Question No : 32 of 33 Mark

Show that any subspace (Y, τ_Y) of a first countable space (X, τ) is also first countable.

Answer (Please click here to Add Answer)

100%

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Question No : 12 of 33

A regular T_1 – space is called a

Answer (Please select your correct option)

- no isolated points
- T_2 – space
- T_3 – space
- none

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Question Summary : 1 Attempted Question

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Question No : 11 of 33

A finite set has _____ in a T_1 space

Answer (Please select your correct option)

- no isolated points
- no limit points
- infinte limit points
- none

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Question Summary : (Attempted Question)

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Question No : 4 of 33

Which of the following statement is false?

Answer (Please select your correct option)

- Discrete topology on a countable set X is second countable.
- Any finite set with any topology is second countable
- Discrete topology on a real line R is second countable
- The set $\mathbb{R}$ with usual topology is second countable

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Question Summary : (Attempted Question)

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Question No : 5 of 33

For metric d on X, an open ball of radius r with center $x \in X$ is defined as:

Answer (Please select your correct option)

- $B(x,r) = \{y \in X \mid d(x,y) \leq r\}$
- $B(x,r) = \{y \in X \mid d(x,y) > r\}$
- $B(x,r) = \{y \in X \mid d(x,y) < r\}$
- none

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Question Summary : (Attempted Question)

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Question No : 6 of 33 Marks: 1 (Budg

Let $B_1(x,r_1)$ and $B_2(x,r_2)$ be two open balls centered at x with radii r_1 and r_2 respectively. Then the following statement is true

Answer (Please select your correct option)

- $B_1(x,r_1) = B_2(x,r_2)$
- either $B_1(x,r_1) \subset B_2(x,r_2)$ or $B_1(x,r_1) \supset B_2(x,r_2)$
- none

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Question Summary (Attempted (question))

Question No : 8 of 33

Which of the following statement is true?

Answer (Please select your correct option)

- Every separable space is second countable.
- Every second countable space is separable.
- R with lower limit topology is second countable.

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Question Summary : 1 Attempted Question

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Question No : 9 of 33

Which of the following statement is true?

Answer (Please select your correct option)

- A separable metric space is second countable.
- A separable metric space is first countable.
- Every metric space is second countable
- none

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Question Summary : (Attempted Question)

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Question No : 10 of 33

A topological space is a T_1 – space if and only if its each finite subset is

Answer (Please select your correct option)

- a perfect set
- a closed set
- an open set
- none

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Question Summary : 1 Attempted Question)

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Question No : 3 of 13

Which of the following statement is false?

Answer (Please select your correct option)

- The set $\mathbb{R}$ with usual topology is metrizable space
- The set $\mathbb{R}^2$ with usual topology is metrizable space
- indiscrete topology on a set X containing more than one point is metrizable

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Question Summary : 1 Attempted Question

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Question No : 1 of 33

Which of the following statement is true?

Answer (Please select your correct option)

- Projection map may or may not be closed
- Projection map is open
- Projection map is closed

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Question Summary : 1 Attempted Question

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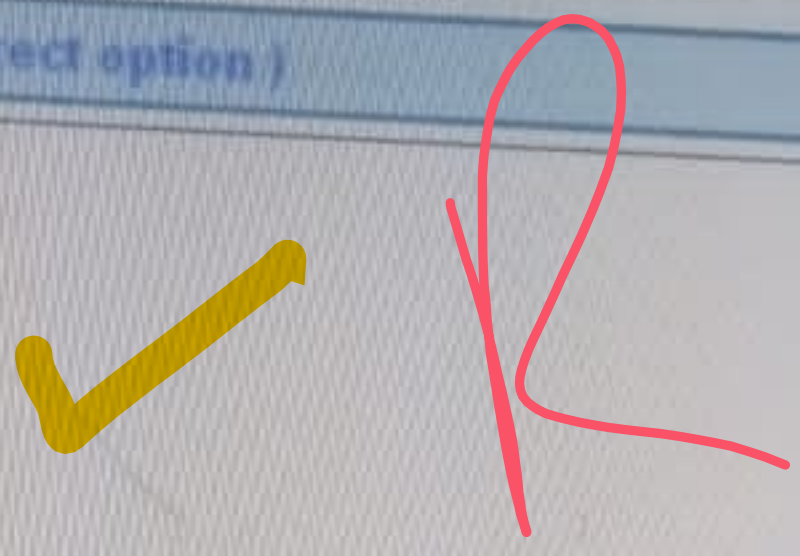
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Question No : 23 of 30

A real line $\mathbb{R}$ with usual topology is path connected.

Answer (Please select your correct option)

- True
- False



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Question Summary : 1 Answered Question

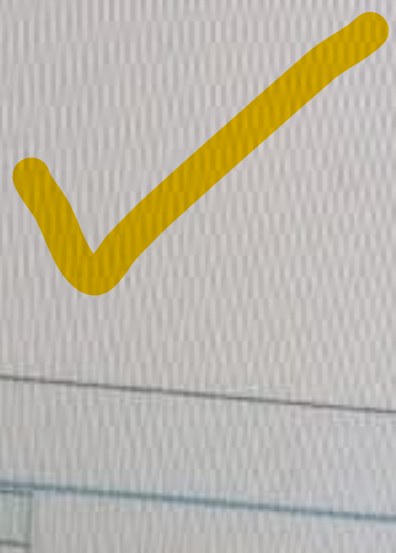
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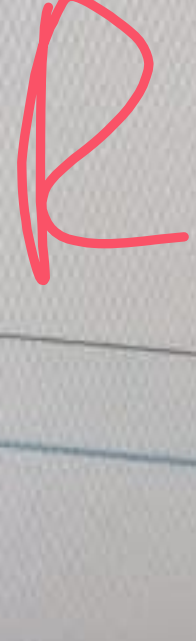
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A discrete space is not locally connected.

Answer (Please select your correct option)

- True
- False





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Question Summary - 1 Attempted Question

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VU Examination Syste...

Question No : 25 of 31

The image of connected topological space is path connected

Answer (Please select your correct option)

- True
- False

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Question Summary : 1 Answered Question

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Question No : 19 of 33

Closure of a connected subset is also connected.

Answer (Please select your correct option)

True

False

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Question No : 20 of 33

Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined as $f(x) = x + 1$. Fixed point of f is:

Answer (Please select your correct option)

- 0
- 1
- 2
- no fixed point

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Question Summary : (Attempted Question)

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Question No : 21 of 33

Let C be a connected component of a topological space X, then

Answer (Please select your correct option)

- C is open
- C is closed
- C is infinite
- none

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Question Summary : (Attempted Question ,)

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Question No : 24 of 33

If (X,τ) be a path connected topological space, then (X,τ) is connected

Answer (Please select your correct option)

- True
- False

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Question Summary - Attempted Question

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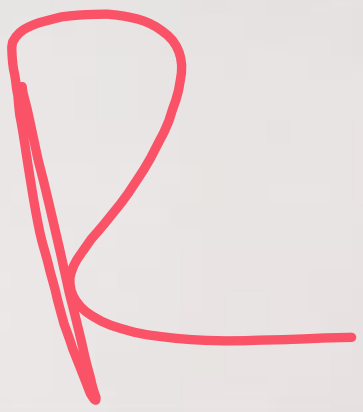
Question No : 16 of 33

Every compact subspace of a hausdorff space is open

Answer (Please select your correct option)

- True
- False





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Question Summary : 1 Attempted Question

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Question No : 27 of 33

Define Normal space.

A topological space (X,T) is said to be complete normal space iff every subspace of X with subspace topology is normal.

Answer (Please click here to Add Answer)

In topology and related branches of mathematics, a normal space is a topological space X that satisfies Axiom T_4: every two disjoint closed sets of X have disjoint open neighborhoods. A normal Hausdorff space is also called a T_4 space

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Question Summary (Attempted Question)

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Question No : 26 of 33

Define metric topology.

Let X be a non empty set with metric d. The topology T on X generated by the set of all open balls in X with respect to d is called metric topology

Answer (Please click here to Add Answer)

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Question Summary : 1 Attempted Question

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MTH634 Topology

Question No : 30 of 33

Show that a set X with topology T containing finite number subsets of X is compact.

Answer (Please click here to Add Answer)

A set X with topology T containing finite number subsets of X is compact.

A finite set X with topology T is compact.

A finite set X with discrete topology T is not compact.

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Question No : 31 of 33

Define connected set

A subset A of a topological space (X,T) is said to be connected if and only if there exist no pair of non empty open subset of U and V of X such that $A \cap U$ and $A \cap V$ are non empty disjoint sets and

$A=(A \cap U) \cup (A \cap V)$

Answer (Please click here to Add Answer)

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MTH634 Topology

Question No : 29 of 33

Consider the sets $X = (1,2,3)$ and $Y = (a,b,c)$ with topologies $\tau_X = (\phi,(1),(1,2),X)$ and $\tau_Y = (\phi,(a,b),(b),Y)$ repectively. Define a map $f: X \to Y$ as:

$f(1) = f(2) = a, f(3) = c,$

Show that f is a closed map.

Answer (Please click here to Add Answer)

100%

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MTH634 Topology

Question No : 28 of 33

Define connected component in a topological space X

Answer (Please click here to Add Answer)

Consider a topological space (X,T), Let $a \in x$ and C be a connected subset of X containing a then

$$C_a = \bigcup_{C \in U} C$$

is called connected component of X containing a.

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Question No : 27 of 33

Define Normal space

A topological space (X,T) is said to be complete normal space iff every subspace of X with subspace topology is normal

Answer (Please click here to Edit Answer)

In topology and related branches of mathematics a normal space is a topologi
two disjoint closed sets of X have disjoint open neighborhood. a normal hausd

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MTH634 Topology

Question No : 26 of 33

Define metric topology.

Let X be a non empty set with metric d. The topology T on X generated by the set of all open balls in X with respect to d is called metric topology.

Answer (Please click here to Edit Answer)

100%

Metric topology: let x be a non empty set with metric d. The topology t or
with respect to d is called metric topology.

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Question No : 33 of 33 Marks: 5 (Budgeted Time Mi

Consider $X = (a,b,c,d,e)$ with topology $\tau = (\phi,(a),(a,b),(a,b,c),(a,d,e),(a,b,d,e),X)$ Show that the set $A = (c,e)$ is disconnected subset of X

Answer (Please click here to Add Answer)

Closure of a connected set is always connected.
Suppose E = A ∪ B, where A ∩ B = ∅ and A ∩ B = ∅,
we show that E is connected by proving that either A or B must be empty. A = A ∩ (A ∪ B) = A ∩ E ⊆ A ∩ B = ∅, which implies that E is connected